International Journal of Applied Earth Observation and Geoinformation 147 (2026) 105196

Contents lists available at ScienceDirect

International Journal of Applied Earth Observation and Geoinformation

journal homepage: www.elsevier.com/locate/jag

Differential land deformation patterns and response mechanism to groundwater recovery in the Beijing-Tianjin-Hebei Plain revealed by reconstructing and clustering of TS-InSAR observations

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ARTICLE INFO

Keywords:
TS-InSAR
K-means
Differential deformation patterns
Groundwater recovery
Beijing-Tianjin-Hebei

ABSTRACT

The Beijing-Tianjin-Hebei (BTH) Plain in north China has experienced significant land subsidence due to long-term groundwater overexploitation. Although water diversion project and climate change have improved groundwater conditions overall, the spatiotemporal heterogeneity in land deformation responses to groundwater recovery remains unclear. This study reconstructs and identifies differentiated deformation patterns across the BTH Plain under the new hydrological background, revealing their multi-scale response mechanisms to groundwater change. After mosaicking Time-Series Interferometric Synthetic Aperture Radar (TS-InSAR) deformation fields, we reconstructed the seasonal and long-term components of deformation. A long-term deformation characteristic-constrained K-means clustering method was developed to classify distinct deformation evolution patterns. Coupling these patterns with groundwater-level (GWL) observations to further distinguish the annual lag and long-term delayed response of deformation to GWL changes. From 2016 to 2024, the BTH Plain exhibits clear spatial partitioning of deformation behaviors, categorized as 10.14% continuous subsidence, 18.86% reduced subsidence, 46.09% stable deformation, and 24.91% slight rebound. Uncertainties in the reconstructed deformation reflect short-term fluctuations driven by extreme hydrological or anthropogenic disturbances. Short-time Fourier transform analysis reveals periodic response of deformation to GWL change and annual lag: elastic deformation dominated areas lag GWL by $\leq$ 2 months, whereas clay-rich persistent subsiding areas lag 4–6 months. The exponential decay model further indicates that the recovery of subsidence funnels is governed by geological structures: funnels intersecting faults show the slowest subsidence decay along their edges, while those without faults display a more uniform edge-to-center recovery pattern.

1. Introduction

Land subsidence has become a major geological and environmental issue worldwide, primarily driven by aquifer system compaction induced by groundwater overexploitation (Galloway et al., 1998, Famiglietti, 2014). Currently, varying degrees of subsidence have been observed in over 50 countries, including California's Central Valley in the United States (Miller et al., 2020), Jakarta in Indonesia (Wu et al., 2022), Tehran in Iran (Haghshenas Haghighi and Motagh, 2024), and the North China Plain (Li et al., 2025, Bai et al., 2022).

The Beijing-Tianjin-Hebei (BTH) region in northern China is one of the most water-scarce areas in the country. Intensive groundwater exploitation over the past decades has led to multiple groundwater funnels across the plain area and triggered significant land subsidence, with maximum cumulative subsidence exceeding 2.7 m (Sun et al., 2022). To alleviate the water resource shortage in northern China, the Middle Route of the South-to-North Water Diversion Project (SNWDP) was officially launched in 2014. According to government reports, more than 70 billion m^3 of water had been delivered through the middle route by February 2025(China South-to-North Water Diversion Corporation,

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https://doi.org/10.1016/j.jag.2026.105196
Received 15 December 2025; Received in revised form 7 February 2026; Accepted 18 February 2026
Available online 21 February 2026
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2025). Benefiting from SNWDP, groundwater management, and frequent extreme precipitation events in recent years, groundwater levels in Beijing and Tianjin have shown notable recovery since 2016 (Lai et al., 2024; Su et al., 2023). However, in some areas of Hebei Province, due to the high demand for agricultural irrigation, the situation of groundwater overexploitation has not been completely reversed (Yang et al., 2021; Yang et al., 2024).

Under this new hydrological background, defined here as the post-SNWDP period characterized by reduced groundwater abstraction and overall groundwater-level recovery conditions, the spatial heterogeneity of land deformation evolution and its multi-scale responses to groundwater-level variations across the BTH Plain remain unclear. This poses a critical challenge for the precise prevention and control of land subsidence at the regional scale.

Interferometric Synthetic Aperture Radar (InSAR) has become a powerful tool for regional scale, high-precision deformation monitoring, with accuracy reaching the millimeter level (Ferretti et al., 2000; Hooper et al., 2004; Fu et al., 2025). The advancement of time-series InSAR (TS-InSAR) techniques (Berardino et al., 2002; Rabus et al., 2004; Ferretti et al., 2011) provides a solid foundation for separating deformation components and efficiently extracting seasonal and long-term deformation trends (Li et al., 2025). When combined with signal decomposition and clustering methods, TS-InSAR can further reveal the spatial heterogeneity of aquifer-system deformation (Chaussard et al., 2014; Bonì et al., 2016), providing a more comprehensive perspective for understanding the physical response mechanism of groundwater and land deformation.

Among the methods of pattern identification, Principal Component Analysis (PCA) enables dimensionality reduction and extraction of dominant modes from the deformation time series of a large amount of high-coherence InSAR points (Wang et al., 2022). Furthermore, Independent Component Analysis (ICA) aims to separate statistically independent signals from mixtures (Peng et al., 2024; Lai et al., 2024). However, its assumptions of linear mixing and statistical independence often limit interpretability in regions with large spatial extents and complex aquifer structures. In contrast, the K-means clustering algorithm does not rely on independence assumptions and has been widely applied in TS-InSAR studies to identify areas with differentiated deformation behavior, such as distinguishing loading-induced response from pore elastic responses in the aquifer system (Kang & Knight, 2023; Liang et al.,2025) or detecting non-stationary deformation zones when combined with PCA (Festa et al., 2023). On this basis, combining clear characteristic parameters of reflecting deformation trend and the K-means algorithm can further improve the interpretability of clustering results, enabling unsupervised identification and physical interpretation of distinct deformation patterns across large regions.

Although SNWDP has been implemented for nearly a decade, its comprehensive impact on regional aquifer restoration and land deformation evolution remains largely unexplored. Existing studies have primarily focused on the single subsidence pattern in typical cities or local areas (Bai et al., 2022; Xu et al., 2025; Li et al.,2025; Han et al., 2023). As a result, how different regions across the BTH Plain respond heterogeneously to groundwater recovery in terms of long-term deformation evolution and short-term variability remains poorly understood at the regional scale. In addition, extreme precipitation has increased in recent years (Kuang et al., 2024), yet its heterogeneous impacts on regional aquifer recharge and deformation responses remain poorly evaluated. Addressing these knowledge gaps can help deepen our understanding of aquifer deformation dynamics under the new hydrological background.

In this study, we perform TS-InSAR processing on Sentinel-1A SAR imagery acquired from 2016 to 2024 to generate a mosaicked deformation field covering the entire BTH Plain. Seasonal and long-term deformation components are reconstructed using harmonic functions and quadratic polynomials, respectively. We further develop a characteristic-constrained K-means clustering framework to identify multiple long-term deformation evolution patterns. Reconstruction uncertainty is evaluated, and the multi-scale responses of deformation to groundwater-level variations, including annual phase lag and long-term delayed compaction, are systematically analyzed among different clusters. By establishing this integrated framework, this study aims to reveal the mechanisms underlying differentiated deformation evolution patterns and their heterogeneous responses to groundwater recovery, thereby providing scientific insights for sustainable groundwater management in regions affected by long-term groundwater overexploitation.

2. Study area and datasets

2.1. Overview of the study area

As shown in Fig. 1, the BTH Plain is the main part of the North China Plain, bordering the Yanshan, Taihang Mountains, and Bohai Sea. It covers the plain areas of Beijing, Tianjin, and Hebei (Fig. 1a), with an area of approximately 87836.67 km^2. The formation of BTH plain area is closely related to the alluvial deposits and geological structures of the Yellow River and the Haihe River. With the continuous accumulation of Quaternary loose sediments, the thickness of the resulting pore aquifer system exceeds 100 m (Zhang et al., 2009).

According to the geomorphic characteristics, BTH covers three parts of the North China Plain from the mountain front to the coast, namely, the piedmont plain (PP), the central plain (CP), and the eastern plain (EP) (Fig. 1b). As shown in Fig. 1c, the single-layer sandy aquifer is mainly distributed in PP, mainly composed of gravel layers mixed with cohesive soil layers. The multi-layered structure is mainly distributed at the bottom of each sub plain, with alternating sand and clay layers vertically. From PP to EP, the thickness of the aquifer gradually decreases, the particle size gradually becomes finer, and the permeability gradually deteriorates (Zhang et al., 2009). In plain areas, warm and humid air currents in summer are blocked by the Taihang Mountains and Yanshan Mountains, allowing rainfall to form easily on the windward slopes. On average, it is about 550 mm per year. The precipitation during the flood season accounts for over 75% of the annual precipitation. Precipitation exhibits significant interannual variability.

2.2. SAR datasets

SAR imagery from three Sentinel-1A acquisition paths covering the BTH Plain from 2016 to 2024 was obtained to investigate the long-term evolution patterns and responses of land deformation. Detailed information on the datasets is provided in Table 1.

2.3. Groundwater-level and precipitation

Groundwater-level (GWL) monitoring was obtained from the China Groundwater Statistics Yearbook compiled by the China Institute of Geo-environmental Monitoring (2018–2024). The datasets provide monthly GWL observations from 2018 to 2024. According to well depth, the collected observations were divided into shallow confined wells (120–210 m) and deep confined wells (250–310 m) (Zhang et al.,2009). The spatial distribution of the wells and the amplitude of GWL fluctuations are provided in the Supplementary Materials S1.

Precipitation amount were obtained from the ERA5-Land reanalysis product developed by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5-Land provides hourly precipitation estimates at a spatial resolution of $0.1^\circ \times 0.1^\circ$ (~9 km). In this study, hourly precipitation was aggregated to monthly totals to ensure temporal consistency with the groundwater-level observations spanning 2018–2024.

3. Methods

In this study, TS-InSAR was applied to process Sentinel-1A SAR

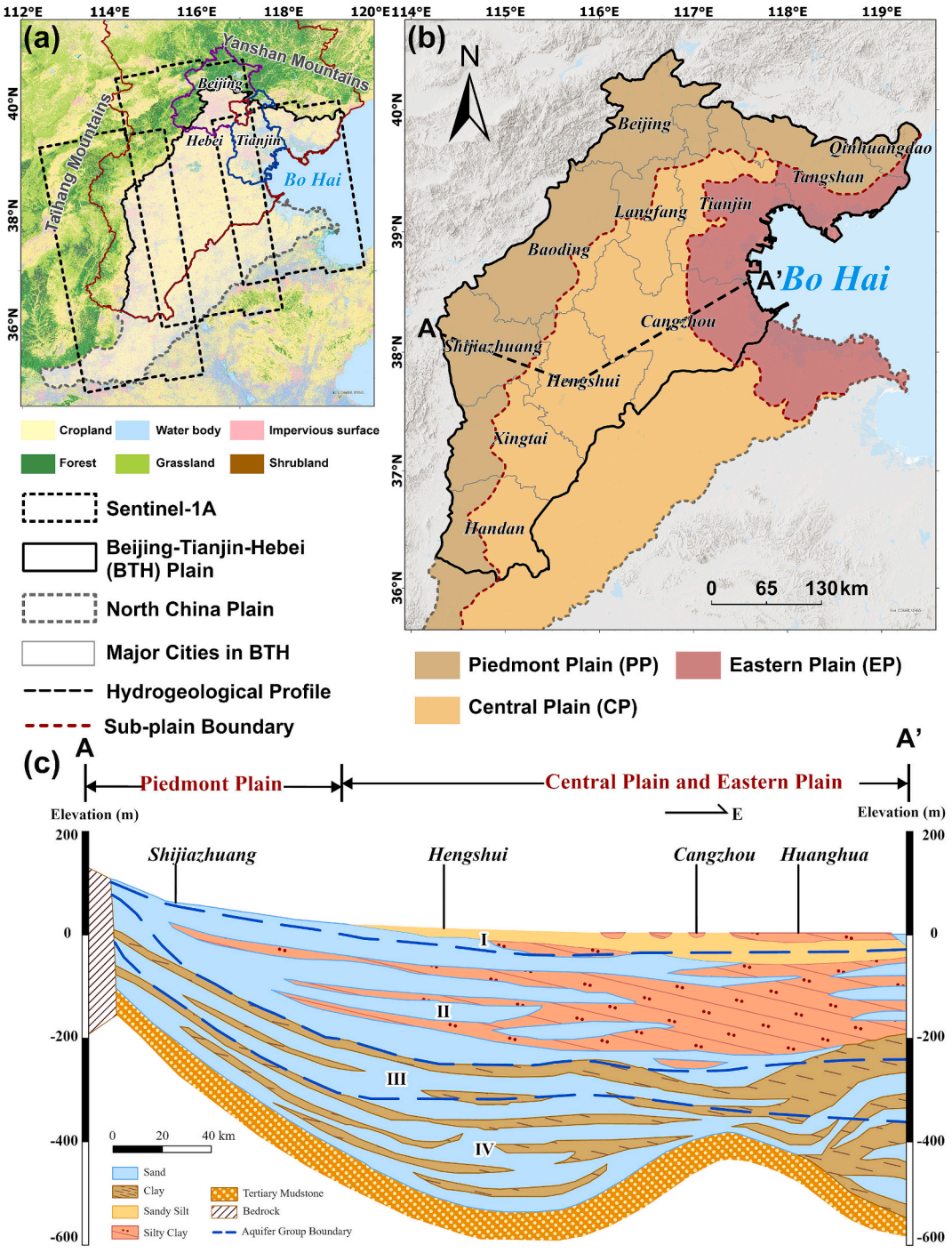

Fig. 1. Overview map of the study area. (a) The location of the Beijing-Tian-Hebei (BTH) Plain. (b) Sub-plain divisions of the BTH Plain. (c) Hydrogeological profile A-A' shown in (b).

Table 1
Sentinel-1A SAR datasets used in this study.

Sensor	Path	Period	Number of images	Direction	Average incident angle
Sentinel-1A	40	2016.01.07 ~ 2024.12.02	154	Ascending	33.7700
	142	2016.01.14 ~ 2024.12.21	147	Ascending	33.7779
	69	2016.01.09 ~ 2024.12.16	122	Ascending	33.7318

imagery to derive deformation time series from 2016 to 2024, and correction and mosaicking of the deformation fields over the BTH Plain were performed. The seasonal and long-term deformation components were reconstructed using a harmonic function and a quadratic polynomial, respectively. A characteristic-constrained K-means clustering approach was then developed to identify different deformation patterns and to evaluate the uncertainty of the reconstructed deformation. Furthermore, this study distinguished the multi-scale delayed response mechanisms of deformation to groundwater level (GWL) variations. The data processing workflow and key steps are illustrated in Fig. 2.

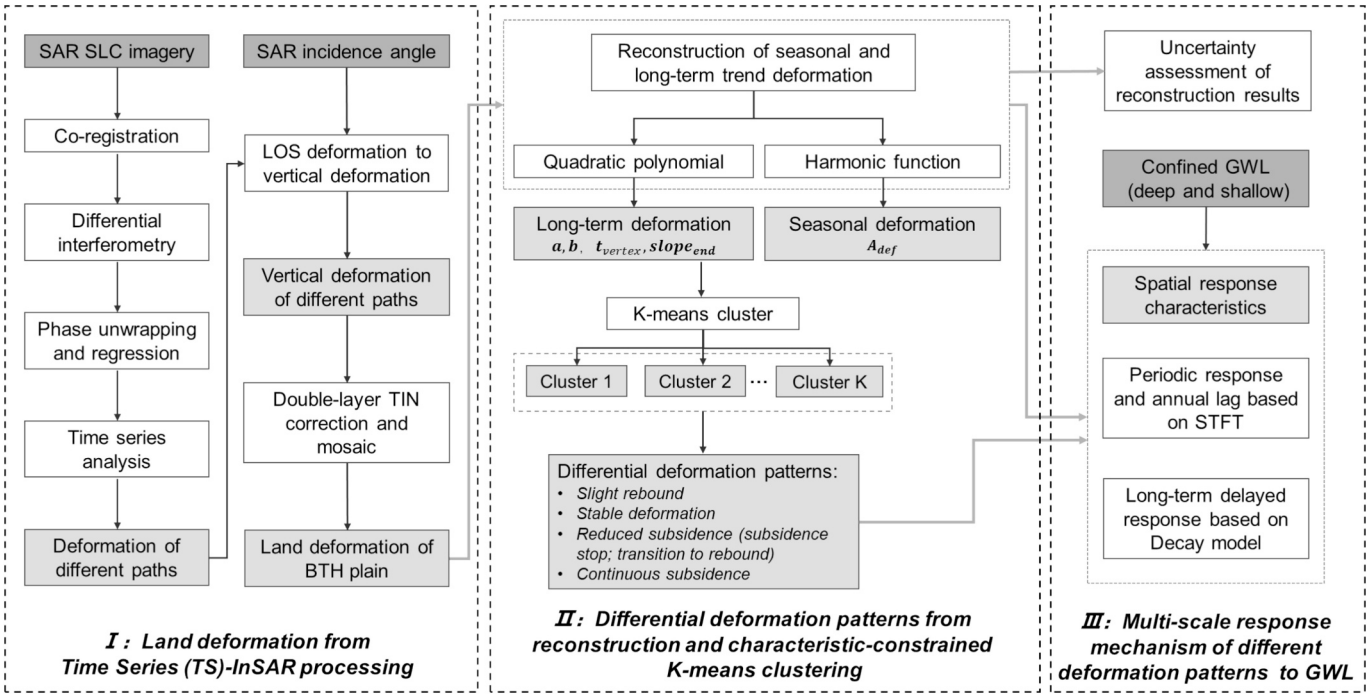

Fig. 2. Data processing workflow and study framework.

3.1. Regional InSAR deformation processing

3.1.1. Time series InSAR analysis

In this study, interferometric point target analysis (IPTA) was used to retrieve land deformation across the BTH Plain from Sentinel-1A imagery acquired along three paths. IPTA iteratively estimates a phase regression model and applies several procedures to reduce topographic residuals and atmospheric effects for more accurate deformation retrieval (Werner et al., 2003, Stramondo et al., 2008).

Sentinel-1A imagery from paths 40, 142, and 69 were concatenated along the azimuth direction. Enhanced spectral diversity was employed to co-register imagery from different paths, where the azimuth co-registration offset was reduced to less than 0.001 pixels to avoid phase jumps during interferometric processing. Azimuth de-ramping was further applied to remove Doppler centroid shifts between adjacent bursts in TOPS mode.

Considering the narrow orbital tube of Sentinel-1, an interferometric network was constructed using combinations of three adjacent scenes. During differential interferogram generation, multi-looking factors of 10×2 in range and azimuth were applied to reduce spatial noise. The 30 m SRTMv3 was used to remove the topographic phase. Adaptive filtering was subsequently applied using window sizes of 64, 32, and 16 pixels in iterative steps to reduce random noise (Xue et al., 2014). Minimum cost flow was selected for phase unwrapping. High-coherence pixels were selected using an average coherence of > 0.6 and a standard deviation of < 0.3, with an additional minimum intensity threshold of 0.01 to suppress water-body and shadow effects, the coherence distribution histograms show in Supplementary Materials S2 (Wang et al., 2022). Unwrapped phases on these pixels were extracted, and singular value decomposition was applied to convert the multi-master interferometric phases into time-series deformation. In the IPTA workflow, atmospheric delays and residual topographic components were iteratively estimated and removed based on their spatiotemporal frequency and baseline dependence. Finally, deformation results were generated for paths 40, 142, and 69.

3.1.2. Correction and mosaic of adjacent path TS-InSAR deformation

To obtain the regional surface deformation field over the entire BTH Plain, the inconsistencies among adjacent InSAR paths must be corrected before mosaicking. In this study, Path 142, which was independently validated using leveling measurements (Fig. 3a), was selected as the reference path.

Horizontal deformation in BTH Plain was considered negligible due to its small magnitude. (Zhang et al., 2026), the line-of-sight (LOS) deformation (d_{los}) measurements of each path were first converted into vertical deformation (d_{ver}) using the incidence angle (θ):

$$d_{ver} = \frac{d_{los}}{cos\theta} \tag{1}$$

After unifying the spatial reference frame, search for the homonymous pixels in the reference and target paths within a spatial distance threshold of 0.5 pixels in the overlapping area (Fig. 3b). Subsequently, the double-layer Delaunay triangulation network (TIN) (Wang et al., 2021; Dong, 2023) was employed to evaluate inter-path consistency and estimate datum offsets. For the reference and target paths, independent Delaunay triangulations were constructed based on the homologous pixels (Fig. 3c).

For each corresponding edge in the triangulation networks, the deformation difference ΔD_{diff} between the two paths was calculated (Fig. 3d), which reflects the consistency of local deformation gradients between adjacent paths. If the local deformation structures are consistent, the inter-path deformation discrepancy can be simplified as a constant vertical datum offset. The standard deviation ΔD_{diff_std} of the triangulation edge differences in the overlapping areas between the reference and target paths remains stable across all months, as in Supplementary Materials S5.

Therefore, in our study, the vertical datum offset between the reference and target paths was estimated as the median of the vertical deformation differences Δv_{median} of homologous pixels in the overlapping area. The median ensures robustness against outliers caused by residual atmospheric effects or local decorrelation. Then, it was uniformly applied to the target path to achieve vertical datum correction. By repeating the above process for adjacent paths of InSAR deformation in all months, the deformation results of Path 40 and Path 69 are unified onto a vertical reference consistent with Path 142, enabling seamless mosaicking and the construction of a spatially continuous TS-InSAR

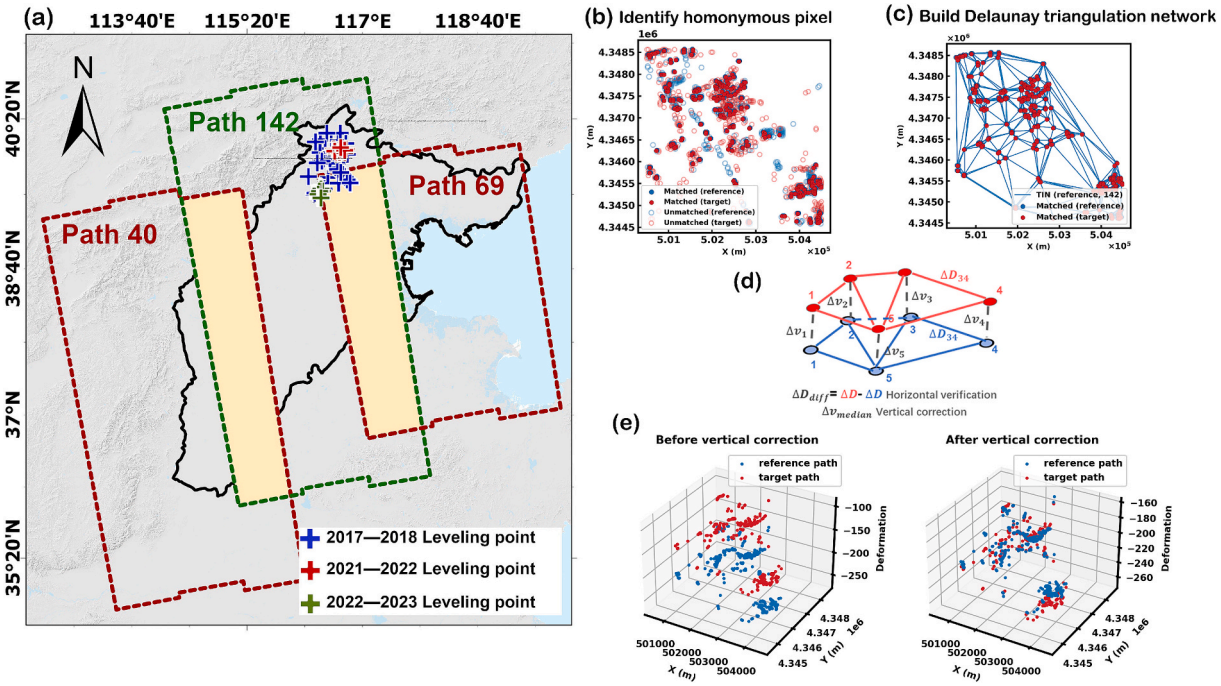

Fig. 3. Schematic diagram of double-layer Delaunay triangulation construction and correction based on overlapping areas (example with an overlapping area of 8.0 $\times$ 10^6 m^2). (a) Spatial distribution of Sentinel-1A ascending paths over the study area and the locations of leveling points. (b) Identification of homonymous pixels between reference and target paths. (c) Construction of the Delaunay triangulation network based on matched homonymous pixels. (d) Verification based on horizontal edge difference and correction based on vertical deviation median. (e) Schematic of before and after correction effects.

deformation field over the BTH Plain.

3.2. Reconstruction of seasonal and long-term trend deformation

3.2.1. Seasonal deformation

Seasonal deformation is primarily driven by the annual cycle of groundwater recharge and discharge. The associated elastic rebound and compaction responses mainly occur within the aquifer system (Neely et al., 2021). To model this periodic behavior, we use a harmonic function (Tang et al., 2023), which only considers the annual component for seasonal deformation in this study:

$$Def_{seasonal}(t) = Acos(2\pi t) + Bsin(2\pi t) \tag{2}$$

$$A_{def} = \sqrt{A^2 + B^2} \tag{3}$$

A_{def} represents the annual amplitude of seasonal deformation.

3.2.2. Long-term trend deformation

Considering that land subsidence in the BTH Plain has undergone multiple evolutionary stages under groundwater exploitation and regulation, the long-term trend deformation component was modeled using a quadratic polynomial fitted by the least-squares method (Li et al., 2025). The formulation is expressed as:

$$def_{long-term}(t) = at^2 + bt + c \tag{4}$$

The quadratic model summarizes long-term deformation time series using some physically interpretable parameters. Specifically, quadratic coefficient a represents the curvature of the long-term trend and indicates acceleration ($a < 0$) or deceleration ($a > 0$) of subsidence or uplift. Linear coefficient b represents the overall linear trend of the fitted curve.

Additionally, we also selected two derived parameters to characterize the long-term evolution of deformation. Turning time $t_{vertex} = -\frac{b}{2a}$ identifies the timing at which the deformation trend experiences a potential transition. Terminal slope $slope_{end}$ reflects the deformation trend slope at the end of the observation period.

The fitted coefficients and derived parameters are subsequently combined to construct a multidimensional feature space for unsupervised clustering analysis, allowing for the objective identification of dominant deformation evolution patterns.

3.3. Characteristic-constrained K-means clustering based on long-term deformation

To identify spatial patterns of long-term deformation evolution in the BTH Plain, we applied K-means clustering algorithm to the reconstructed long-term trend deformation parameters derived in Section 3.2.2. The purpose of this approach is to group TS-InSAR results with similar long-term deformation behaviors into distinct clusters, thereby revealing dominant spatial patterns while avoiding the influence caused by the short-term noise in the original time series.

K-means is an iterative, distance-based unsupervised classification method that partitions a dataset into K clusters by minimizing the within-cluster variance (Peng et al., 2024). Specifically, it minimizes the Mean Sum of Euclidean Distances (MSED) between each feature vector and the centroid of its assigned cluster. The algorithm iteratively updates cluster centroids and reassigns samples until convergence, which is defined as the stabilization of centroid positions.

Instead of clustering the original high-dimensional TS-InSAR time series, which are sensitive to short-term fluctuations, we constructed a four-dimensional feature vector for each point to represent key long-term deformation characteristics:

$$x_i = \begin{bmatrix} a_1, b_1, t_{vertex1}, slope_{end1} \\ a_2, b_2, t_{vertex2}, slope_{end2} \\ \cdots \\ a_i, b_i, t_{vertexi}, slope_{endi} \end{bmatrix}, i = 1, 2, 3 \cdots n \tag{5}$$

These features provide a comprehensive, physically interpretable representation of long-term deformation, which is used to construct a multidimensional feature space for K-means clustering. The resulting clusters are later compared with those obtained from clustering the

original time series.

Because the selected long-term deformation features have different physical units and numerical ranges, all features were standardized using z-score normalization before K-means clustering. Specifically, each feature was standardized by subtracting its mean and dividing by its standard deviation, ensuring that all variables contribute equally to the Euclidean distance calculation. This preprocessing step prevents parameters with larger numerical magnitudes from dominating the clustering results.

Initially, cluster centroids are randomly selected. Each sample is then assigned to the nearest centroid based on Euclidean distance:

$$d(x_i, c_k) = \sqrt{\sum_{m=1}^{M} (x_{i,m} - c_{k,m})^2} \tag{6}$$

Where $x_i = [x_{i,1},\ x_{i,2} \cdots x_{i,M}]^T$ is the feature vector of TS-InSAR point i. c_k is the centroid of cluster k. $M = 4$ is the number of features.

After assignment, cluster centroids are updated by minimizing the within-cluster MSED:

$$MSED_k = \frac{1}{N_k} \sum_{x_i \in C_k} d(x_i, c_k) \tag{7}$$

$$c_k^{(t+1)} = \frac{1}{N_k} \sum_{x_i \in C_k} x_i \tag{8}$$

This assignment-update cycle is repeated until convergence or the maximum number of iterations is reached. The optimal number of clusters K is determined using the Elbow Method, which identifies the point at which additional clusters provide minimal reduction in within-cluster sum of squared errors (SSE).

By clustering on these physically interpretable long-term trend characteristic parameters rather than the original TS-InSAR time series, this approach reduces the influence of short-term noise while preserving meaningful and representative deformation patterns.

4. Results and analysis

4.1. Land deformation from TS-InSAR

4.1.1. Validation of TS-InSAR results

To evaluate the reliability of TS-InSAR deformation results, we validated the InSAR deformation results from Sentinel-1A Path 142 using leveling measurements taken in 2017–2018, 2021–2022, and 2022–2023, respectively. As shown in the Fig. 4a. Most of the point errors are distributed within ± 10 mm, confirming the accuracy and stability of Path 142. This result supports its selection as the reference path for correction. Based on the correction using verified reference path, the deformation results of the target Path 69 were further validated using leveling data (Fig. 4b). After applying the vertical datum correction, the agreement between TS-InSAR deformation and leveling observations is significantly improved.

In addition, deformation rate maps before and after mosaicking were visually compared in the Supplementary Materials S5, revealing enhanced spatial continuity across adjacent paths after correction. The monthly deformation differences between Path 142 and its adjacent Path 40 and Path 69 were quantified before and after correction, and the *RMSE* were consistently reduced. It demonstrates the reliability and applicability of the correction and mosaicking according to Section 3.1.2 for constructing a consistent deformation field over the BTH Plain.

4.1.2. Annual deformation rate

Fig. 5 presents the vertical deformation rates derived from TS-InSAR observations. It was divided into two periods (2016–2018 and 2018–2024) to reflect a transition in groundwater management policy. In 2018, the Ministry of Water Resources of China and the Hebei Provincial Government jointly formulated groundwater replenishment plans, focusing on solving the problem of groundwater overexploitation in North China (Ministry of Water Resources of China, 2018). In addition, continuous and regionally consistent GWL observations became available from 2018 to 2024 in this study, providing a reliable basis for systematically analyzing post-management deformation and groundwater responses.

Significant land subsidence (≥40 mm/yr) is primarily concentrated in the central and eastern plains. These subsiding areas spatially coincide with historical confined groundwater funnels from Guo et al., 2021, indicated by the green and purple dashed outlines, confirming that regional subsidence across the BTH Plain has been predominantly driven by long-term groundwater overexploitation.

A comparison of the two periods reveals a substantial slowing down in subsidence during 2018–2024. The major subsidence funnels gradually shrink, and some have even disappeared. As shown by the deformation rate histograms in Fig. 5, the distribution for 2018–2024 shifts toward less negative values and becomes more concentrated compared to that for 2016–2018, indicating reduced spatial variability of land subsidence. Specifically, the 0.1%–99.9% range of the deformation rate narrows from −112.1–28.0 mm/yr (2016–2018) to −98.4–26.5 mm/yr (2018–2024), reflecting a notable slowdown of ground subsidence in response to improved groundwater conditions.

4.2. Seasonal and long-term deformation components reconstructed from TS-InSAR

4.2.1. Seasonal deformation amplitude

Fig. 6a shows the spatial distribution of seasonal deformation in the

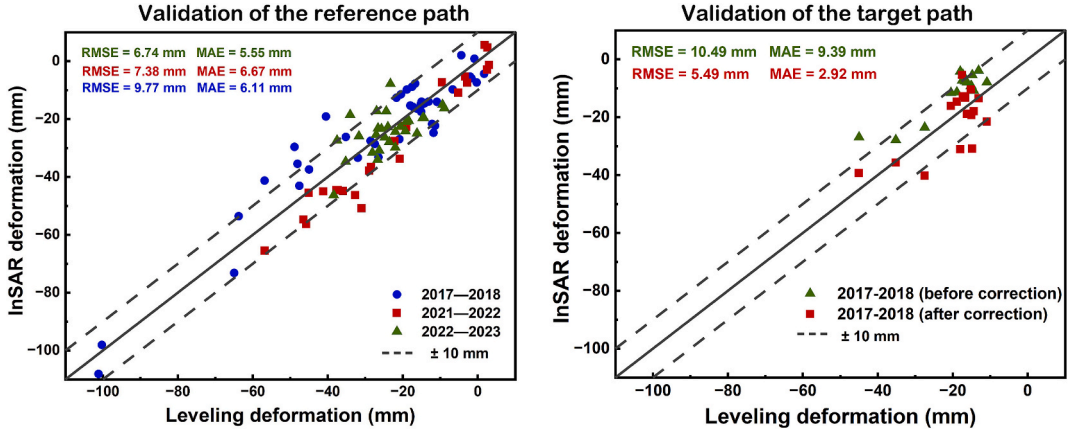

Fig. 4. Accuracy validation for TS-InSAR deformation with leveling.

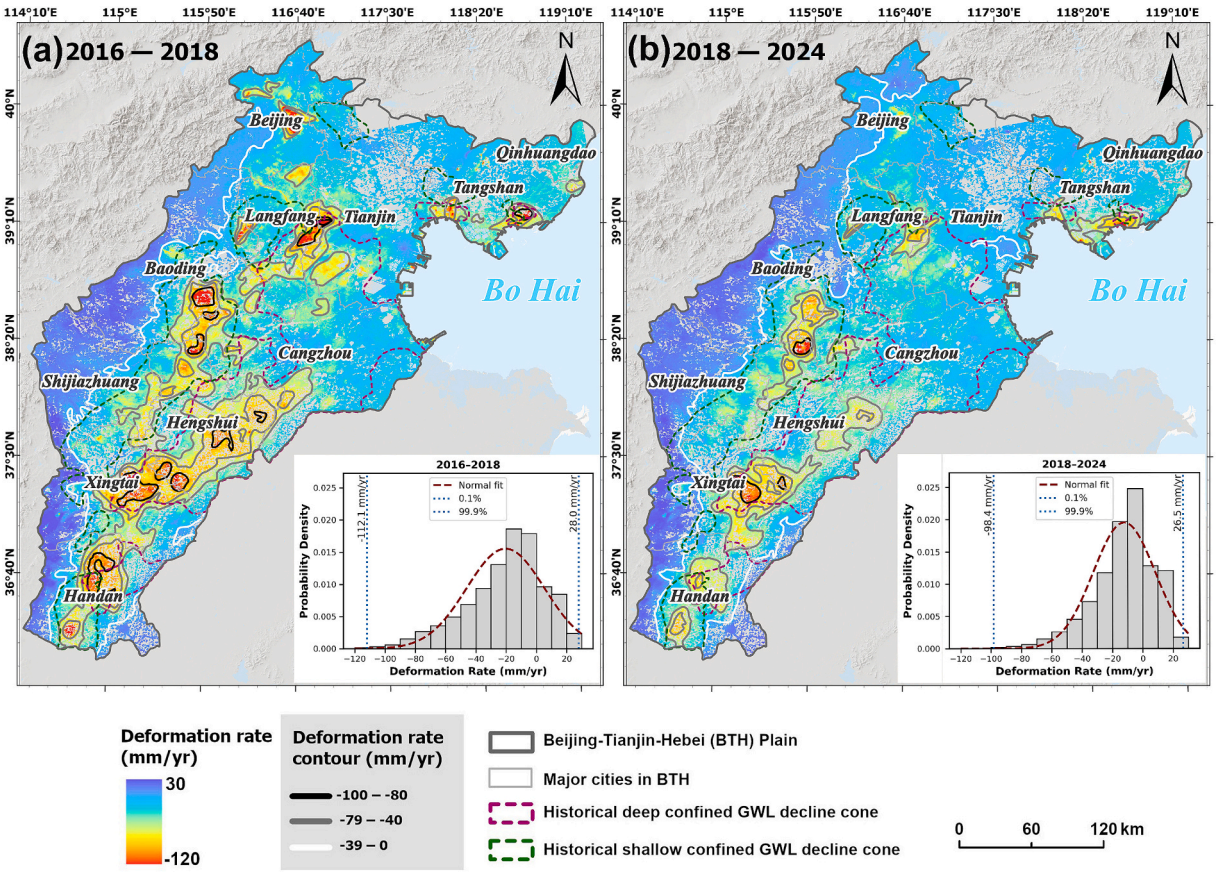

Fig. 5. The annual deformation rate from TS-InSAR and Sentinel-1A in the Beijing-Tianjin-Hebei (BTH) Plain. The scope of historical groundwater funnels is from Guo et al., 2021. Subgraph shows histograms of deformation rates for each period, with dashed curves indicating normal distribution fits and vertical dashed lines marking the 0.1% and 99.9% quantiles. (a) 2016–2018. (b) 2018–2024.

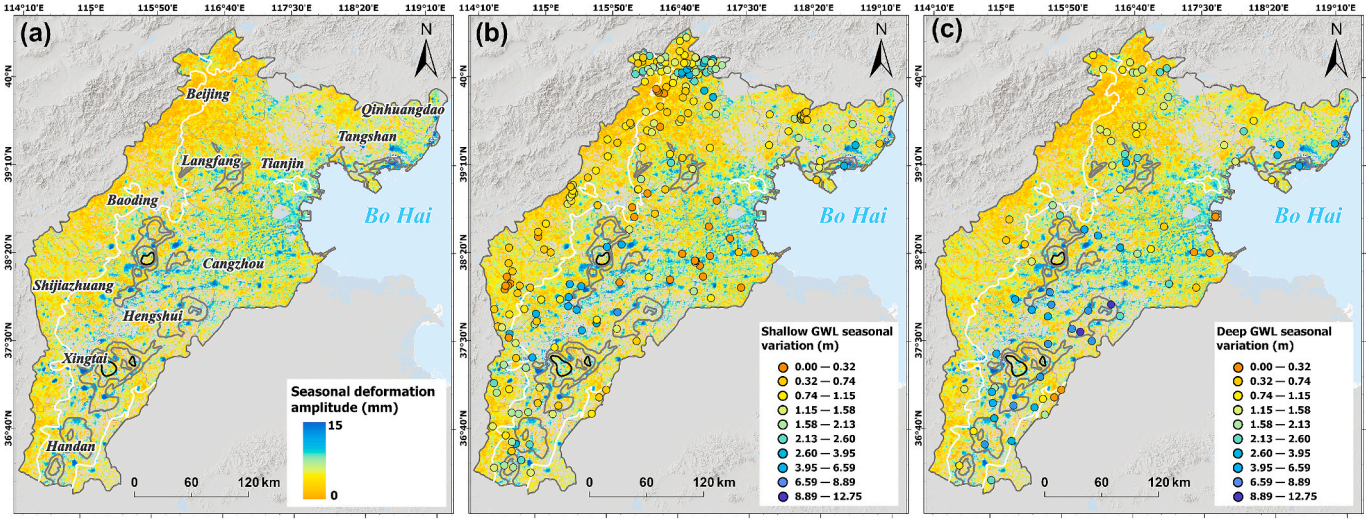

Fig. 6. Seasonal deformation reconstructed based on harmonic function. (a) spatial distribution (b) comparison with GWL seasonal variation of shallow confined wells monitoring (c) comparison with GWL seasonal variation of deep confined wells monitoring.

BTH Plain reconstructed based on the harmonic function, with the annual amplitude A_{def} ranging from 0 to 15 mm. The piedmont zone exhibits relatively weak seasonal deformation, generally below 5 mm. In contrast, pronounced seasonal deformation is concentrated in the central and eastern plains, especially in major subsidence funnels. The results reveal a strong spatial correspondence between the seasonal amplitudes of GWL fluctuations and seasonal deformation, indicating that the pronounced seasonal deformation in the central and eastern plains is primarily governed by periodic aquifer recharge and discharge.

As illustrated by the hydrogeological profile in Fig. 1c, the transition from the piedmont to the central and eastern plains is characterized by increasingly interbedded clay layers and a more extensive distribution of confined aquifers, which increases the aquifer response to groundwater pumping. More importantly, Fig. 6b and Fig. 6c show that the seasonal

GWL amplitudes from deep confined wells are generally larger than those in shallow wells; this effect is particularly pronounced around the Xingtai and Hengshui subsidence funnels. This suggests that deep groundwater extraction and associated variations in aquifer hydraulic heads may play a dominant role in driving the seasonal elastic deformation in these areas.

4.2.2. Long-term deformation characteristics derived from quadratic fitting

Fig. 7 presents the spatial distributions and kernel density estimation (KDE) of the four long-term deformation characteristic parameters, including the curvature coefficient a, the linear coefficient b, the vertex time t_{vertex}, and the slope of the terminal period $slope_{end}$. All parameters exhibit heterogeneous spatial patterns and broad statistical distributions, indicating the variability in long-term deformation characteristics across the plain.

The curvature coefficient a, which reflects deformation acceleration, and the KDE exhibits a right-skewed distribution, with most values concentrated at low to moderate levels and a small number of high values and negative values. It indicates different degrees of deformation deceleration in specific regions. The linear coefficient b represents the overall long-term deformation linear trend. The KDE displays a wider and more peak. Its spatial pattern exhibits stronger regional continuity, and high values correspond to the distribution of settling funnels compared with a. The time for the transformation trend to change is represented by the x-axis value of the extremum point of the quadratic curve. Therefore, $t_{vertex} < 0$ indicates that the deformation trend transition occurred earlier than our monitoring period, while $t_{vertex} > 96$ months indicates that the transition has not yet occurred during the monitoring period. Most values occur within the monitoring period or early months from KDE, while only a small portion extends to later times. This suggests that deformation transitions generally occur early, whereas delayed evolution and continuous subsidence are limited to certain areas. The pronounced spatial heterogeneity of t_{vertex} highlights significant regional differences in the change of the deformation stage. The end slope $slope_{end}$ shows a wide distribution with both positive and negative values, and its spatial pattern differs from that of b, indicating that post-transition deformation does not simply continue the initial trend but may evolve to different directions.

These parameters capture complementary characteristics of deformation intensity, trend, and timing, forming a comprehensive and interpretable feature space suitable for identifying deformation evolution patterns through unsupervised clustering.

4.3. Differential deformation patterns identified by characteristic-constrained K-means clustering

4.3.1. Clustering results and spatial distribution of deformation patterns

According to the elbow method (Fig. 8a), the whole deformation field of BTH Plain is classified into four homogeneous subregions ($k = 4$). Fig. 8b illustrates the distribution of the four clusters in the standardized feature space defined by the quadratic long-term deformation parameters. The three-dimensional scatter plot constructed using a, t_{vertex}, and $slope_{end}$, with color indicating the standardized linear coefficient b, reveals a clear separation among the four clusters. The selected features effectively capture distinct long-term deformation behaviors.

As shown in Fig. 9, clusters with similar long-term deformation characteristics often form relatively continuous regions, which supports the robustness of clustering results. Cluster-3 (18.86%) and cluster-4 (10.14%) represent the areas experiencing the most severe subsidence in the plain, mainly located within and around the major subsidence funnels. These two clusters are characterized by small and negative b values, indicating intense subsidence. However, unlike cluster-4, which exhibits a completely negative $slope_{end}$, cluster-3 has a wider range of $slope_{end}$ values, ranging from weak negative to positive, reflecting a significant deceleration of subsidence at the end of monitoring period (Fig. 9a). Spatially, cluster-3 is distributed in the subsiding areas of Beijing, western Tianjin, eastern Hengshui, and northern Handan. Cluster-2 (46.09%) exhibits relatively small deformation magnitudes. With $slope_{end}$ concentrated around zero and moderate t_{vertex}, indicating a general stabilization trend toward the end of the monitoring period. Cluster-1 (24.91%) is characterized by small a, near-zero or weakly negative b, and predominantly positive $slope_{end}$, suggesting stable to slightly uplifting deformation behavior. This cluster is mainly located in the western and northern piedmont regions, where the aquifer system is structurally more homogeneous (Fig. 1c).

As shown in Fig. 10, the statistical distributions of the long-term deformation parameters further highlight the distinct characteristics of each cluster. Clear differences are observed among clusters in terms of curvature (a), linear rate (b), vertex timing (t_{vertex}), and end-stage deformation rate ($slope_{end}$), indicating differentiated long-term deformation evolution patterns. Notably, Cluster-3 is characterized by larger a than other clusters, and exhibits a broader spread in $slope_{end}$, indicating a more pronounced transition from subsidence to deceleration.

In addition, to demonstrate the rationality of characteristic-constrained K-means clustering, we compared the clustering results in this study with K-means clustering based on the original deformation time series in the Supplementary Materials S3. We found that clustering based on the original time series, which is high-dimensional, produces fragmented spatial patterns with reduced regional continuity. Particularly in major subsidence funnels, there are often multiple clusters

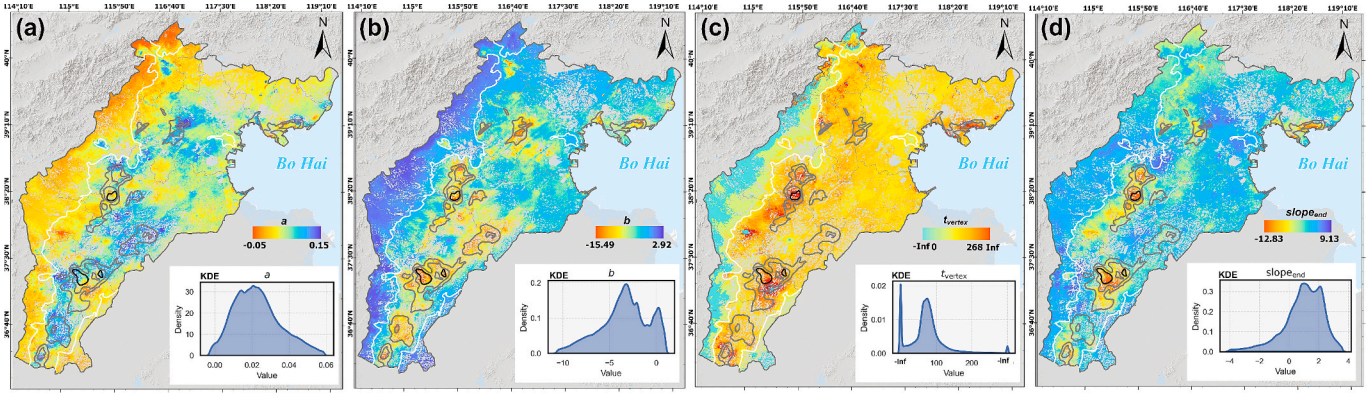

Fig. 7. Input of K-means clustering algorithm, long-term trend deformation characteristic parameters. Spatial distributions of (a) curvature coefficient a (b) Linear coefficient b. (c) Vertex time t_{vertex}. (d) Slope of the terminal period $slope_{end}$. Subgraphs show kernel density estimation (KDE) curves of each parameter, illustrating their statistical distributions.

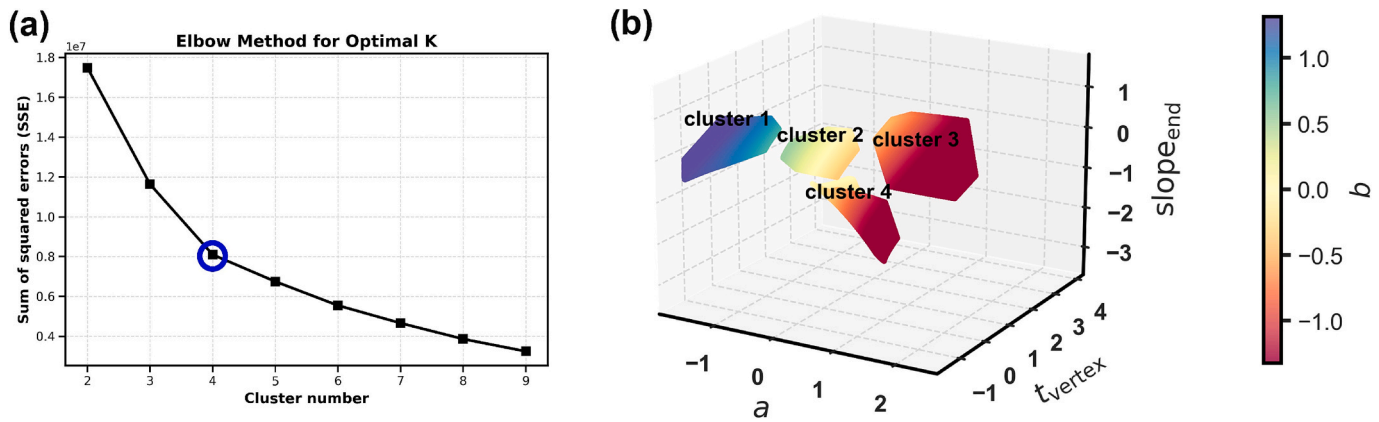

Fig. 8. Clustering strategy of the deformation field in the BTH Plain based on characteristic-constrained K-means. (a) Determination of the optimal number of clusters using the elbow method, where k = 4 is selected. (b) Distribution of the four clusters in the standardized feature space defined by the quadratic deformation parameters.

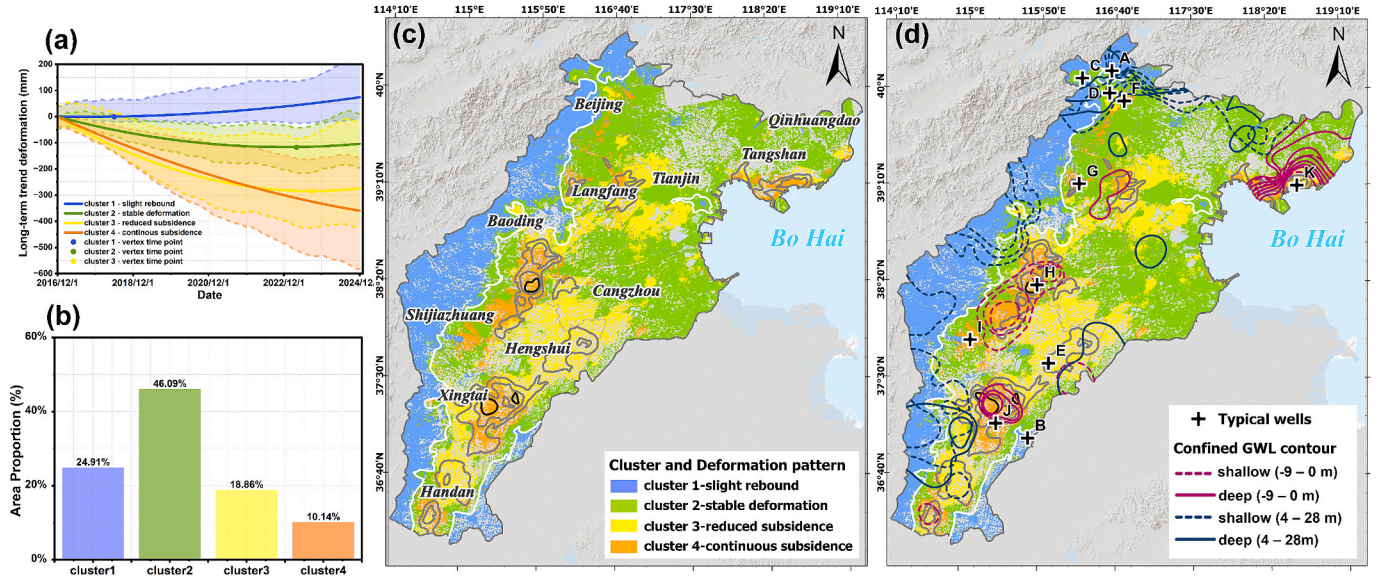

Fig. 9. Deformation pattern clusters from characteristic-constrained K-means clustering (a) Long-term deformation evolution pattern, with shaded areas indicating the variability within each cluster. (b) Area proportion. (c) Spatial distribution. (d) Comparison between clustering patterns and decline contour of deep and shallow confined GWL.

distributed.

In contrast, the characteristic-constrained clustering reflects the spatial heterogeneity of deformation patterns, with cluster-4 showing a clear spatial correspondence with GWL decline cones at different depths, as shown in Fig. 9d. Specifically, long-term subsidence in Baoding-Hengshui and Handan is mainly associated with shallow confined GWL decline, whereas subsidence in Langfang and Xingtai is predominantly controlled by deep confined GWL decline. In the Tangshan coastal area, subsidence corresponds to concurrent declines in both shallow and deep GWL. This spatial consistency indicates that the reconstructed long-term deformation characteristics effectively capture the integrated response of the aquifer system to groundwater extraction, providing independent physical support for the characteristic-constrained K-means clustering results.

4.3.2. Time-series response of different deformation patterns

To investigate the correlation between land deformation and GWL variations across the plain, representative wells were selected from each cluster. For each well, the corresponding TS-InSAR deformation time series and GWL time series were extracted and compared (Fig. 11).

Wells A and B, both located in cluster-1, exhibit clear rebound behavior, with GWL continuously rising from 2018 to 2024 and correlation coefficients generally exceeding 0.65. In particular, Well A shows an almost perfect temporal correspondence between deformation and GWL variations. Wells C in cluster-2 and D, E located in cluster-3 show that, although deformation does not respond immediately to short-term GWL fluctuations, subsidence rates gradually decrease after 2023, and partial rebound is observed in some cases. This indicates a delayed deformation trend response relative to GWL rises. For wells in cluster-4 (Wells F, G, and H), GWL fluctuates with seasonal mining, and the trend change is not significant, whereas deformation continues to exhibit a subsidence trend throughout the study period. The correlation between deformation and GWL in these wells remains low, suggesting that short-term GWL rises are not directly reflected in the observed deformation signals because of the long-term compaction of clay. Well K presents a contrasting behavior, with continuous subsidence showing a strong positive correlation with declining GWL over the entire period.

Despite the general recovery of groundwater levels across the BTH Plain, distinct deformation response patterns are observed among clusters, ranging from rapid response to delayed or weak coupling between

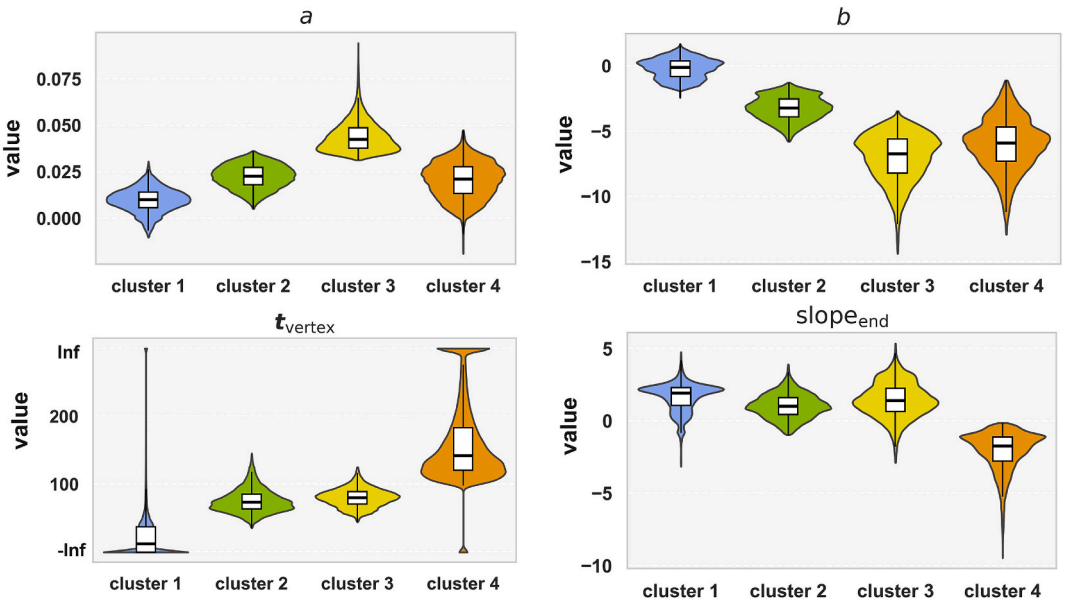

Fig. 10. Statistical distributions of the long-term deformation parameters for the four clusters. The violin plots show the distributions of curvature coefficient *a*, linear coefficient *b*, vertex time t_{vertex}, and slope of the terminal period $slope_{end}$ for each cluster, with the embedded boxplots indicating the median and inter-quartile range.

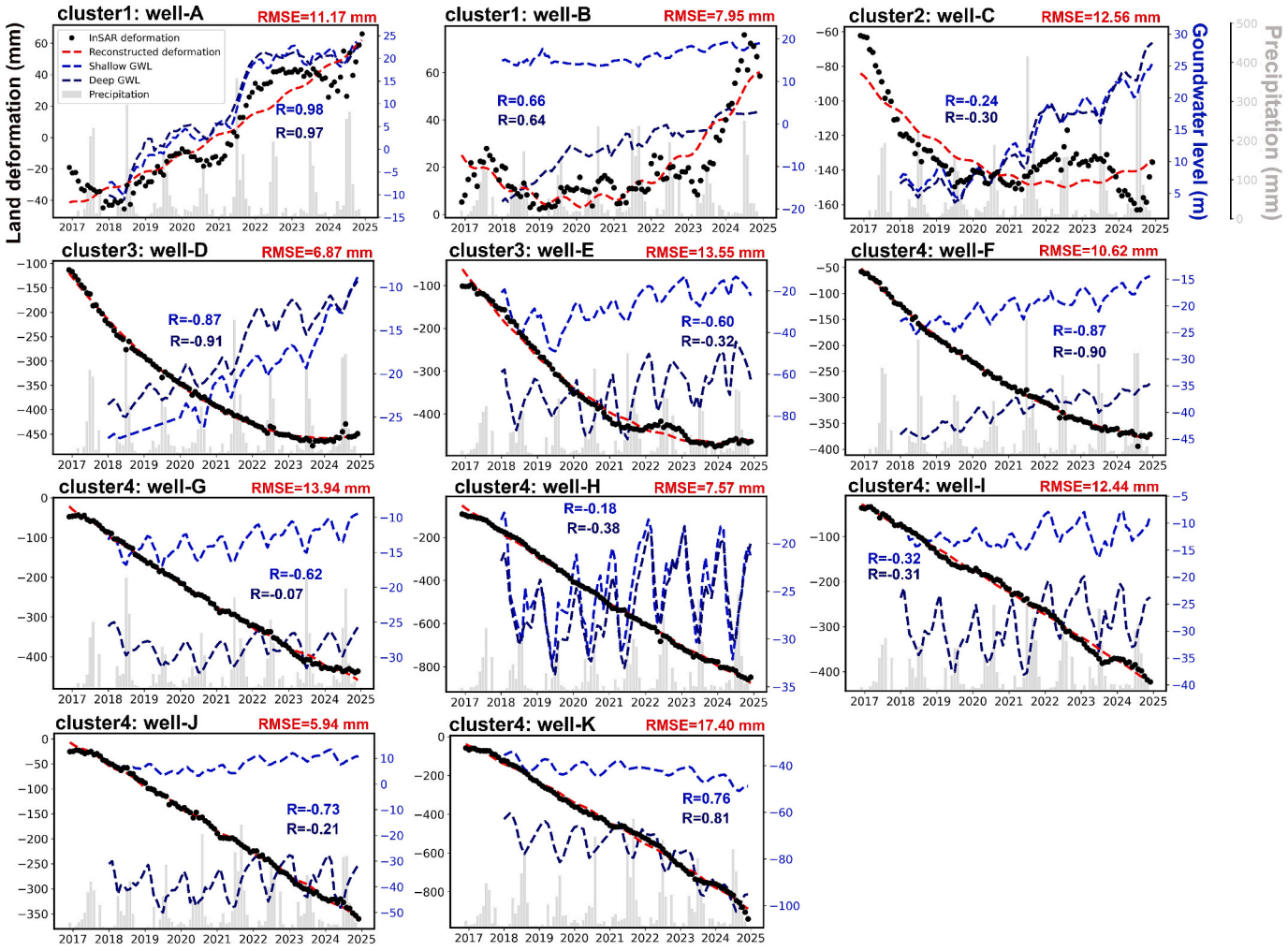

Fig. 11. Comparison of GWL and TS-InSAR deformation time series of 11 wells in different clusters in Fig. 9. Black dots represent TS-InSAR deformation time series. The red dashed line represents the reconstructed deformation. The blue dashed line represents the GWL time series. The gray bar represents monthly precipitation.

deformation and groundwater level changes.

Additionally, comparison between the original TS-InSAR deformation time series and the reconstructed seasonal signals (red dashed lines in Fig. 11) indicates that wells with larger RMSE values tend to exhibit stronger short-term fluctuations. These deviations are particularly evident during specific periods, such as late 2020–early 2022 for Well A and 2022 for Wells C and D. This suggests that, although the seasonal deformation is well captured by the harmonic reconstruction, extreme hydrological events can induce abrupt and short-lived changes in GWL whose deformation responses fall outside the scope of seasonal modeling. Therefore, the residual between reconstructed and original deformation series serves as an indicator of aquifer response to such extreme hydrological or anthropogenic disturbance, helping us further understand the response mechanisms of different deformation patterns.

5. Discussion

5.1. Uncertainty assessment of reconstructing seasonal and long-term trend deformation

To evaluate the performance and uncertainty of the reconstruction model (harmonic function and quadratic polynomial) for TS-InSAR deformation, we calculated the root-mean-square error (*RMSE*) and the normalized RMSE (*NRMSE*). RMSE represents the absolute fitting error of the reconstructed time series and is defined as:

$$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\left(Def_{InSAR}(t) - Def_{model}(t)\right)^2} \tag{9}$$

where $Def_{InSAR}(t)$ and $Def_{model}(t)$ denote the observed and reconstructed deformation at epoch t, and N is the total number of observations.

NRMSE measures the relative error, i.e., *RMSE* normalized by the cumulative deformation:

$$NRMSE = \frac{RMSE}{|Def_{cum}|} \times 100\% \tag{10}$$

with the cumulative deformation defined as $Def_{cum} = Def_{InSAR}(t_N) - Def_{InSAR}(t_1)$.

A comparison between the original deformation time series and the reconstructed components in Section 4.3.2 (red dashed line) indicates that reconstruction uncertainty primarily arises from measurement noise and short-term deformation variability, such as rebound or subsidence induced by extreme hydrological events and abrupt anthropogenic disturbances (e.g., intensive pumping). In this study, the purpose of deformation reconstruction is not to eliminate these short-term signals, but to separate high-frequency variability from the intrinsic seasonal and long-term deformation components. This approach allows localized short-term deformation to be retained in the residuals, while the dominant deformation evolution patterns are robustly characterized.

As shown in Fig. 12, *RMSE* is generally small in the piedmont area but increases significantly in the central and eastern subsidence funnels. In contrast, the spatial pattern of *NRMSE* does not fully coincide with that of *RMSE*. The typical subsiding funnels corresponding to cluster-3 and cluster-4 exhibit consistently low *NRMSE* (all below 6%), indicating that the adopted quadratic polynomial and harmonic functions successfully capture the deformation evolution in these areas. In contrast, cluster-1 and cluster-2, characterized by small deformation magnitudes, include parts of the piedmont and coastal regions that largely contain a higher proportion of noise.

Under conditions where *NRMSE* remains small, *RMSE* still reflects the influence of short-term deformation fluctuations on the reconstruction. For example, extreme rainfall events or abrupt pumping-induced rebound or subsidence can create high-frequency signals in the time series, resulting in high *RMSE* but relatively low *NRMSE*. This indicates that the long-term trend has been well captured, but at the same time, local short-term changes are more pronounced, as shown by the distance fluctuation between the red dashed line and the black dot in the Fig. 11.

5.2. Influence of water-use structure on deformation patterns in the BTH plain

Section 4.3 shows that the characteristic-constrained K-means clustering identifies four dominant long-term deformation patterns across

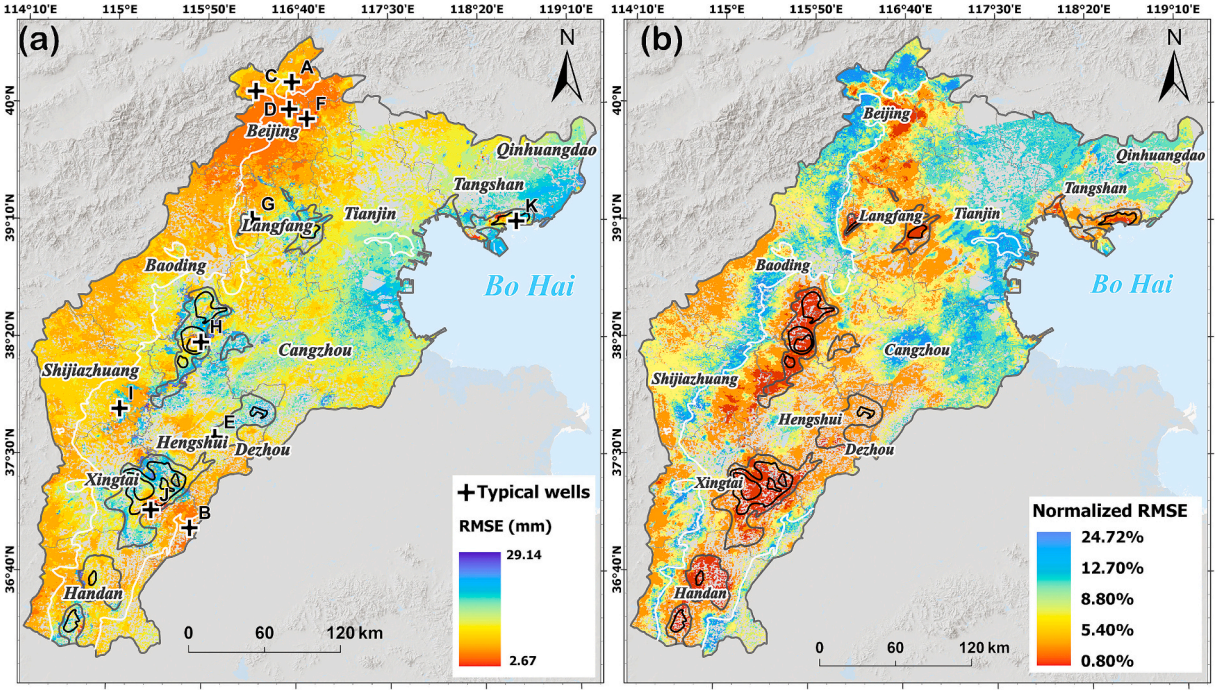

Fig. 12. The spatial distribution of uncertainty indicators for reconstructing deformation. (a) Root Mean Square Error (*RMSE*). (b) Normalized Root Mean Square Error (*NRMSE*).

the BTH Plain, ranging from slight rebound (cluster-1) and stable deformation (cluster-2) to reduced subsidence (cluster-3) and continuous subsidence (cluster-4). These deformation patterns reflect heterogeneous responses of the aquifer system under different groundwater management conditions.

Compared to the other three deformation patterns, cluster-3 exhibits significant transitional characteristics, representing an intermediate stage of evolution from rapid subsidence to other deformation states (Fig. 9). This is because this pattern shows a relatively large linear subsidence rate (parameter *b*), and its terminal slope $slope_{end}$ shows a wide distribution and crosses zero values (Fig. 10), indicating that this cluster does not correspond to a single deformation evolution behavior. Based on the distribution characteristics of $slope_{end}$, we further introduce explanatory segmentation: the areas with $slope_{end}$ higher than the upper quartile of cluster-3 are defined as subsidence to rebound category (cluster-3a), while the remaining areas are classified as stopped subsidence category (cluster-3b), shown in Fig. 13. This subdivision is only used to explain different deformation evolution directions and does not change the original clustering structure.

We collected the proportion of groundwater supply from the Water Resources Bulletins of Beijing, Tianjin, and Hebei (Beijing Water, 2008–2024; Tianjin Water, 2008–2024;Hebei Water, 2008–2024). There are significant differences in groundwater supply structure between Beijing and Tianjin and other cities in Hebei Province. Due to the control of groundwater overexploitation and the impact of SNWDP, the proportion of groundwater supply in Beijing and Tianjin has continued to decline since 2008 (as shown by the dashed line in the Fig. 13c), effectively alleviating the sustained development of land subsidence. Especially in Tianjin, the proportion of groundwater supply remained at the lowest level during the research period. Correspondingly, the cluster-3a that experienced subsidence in the early stage but showed significant rebound in the late monitoring stage is mainly concentrated in Tianjin, which is consistent with the Su et al., 2023. The subsidence area in Beijing is mainly cluster-3b, which is characterized by a significant slowdown in subsidence and tends to stop. This indicates that in the new hydrological background of SNWDP implementation, the sustained decline in groundwater extraction intensity is an important premise for alleviating aquifer compression, rather than just short-term fluctuations in groundwater levels.

Cluster-3a is also distributed in Cangzhou and Hengshui. The proportion of groundwater supply in these two cities has remained relatively low compared to other cities in Hebei Province since 2020. In contrast, cluster-4, which represents continuous subsidence characteristics, is mainly distributed in areas such as Baoding, Xingtai, Handan, and Tangshan. These cities still have a relatively high proportion of groundwater supply around 2020. From the distribution of contour lines of groundwater level decline (Fig. 9d), it can be seen that the aquifer system in these areas is still continuously experiencing an increase in effective stress, thus maintaining a subsidence process dominated by delayed compression.

5.3. Periodic and time-lag response of different deformation patterns to groundwater level changes

In the BTH Plain, long-term subsidence remains the dominant deformation pattern. However, this long-term trend, driven primarily by the cumulative stress associated with historical groundwater overexploitation, tends to obscure the short-period deformation signals linked to current GWL fluctuations and precipitation events. As discussed in Section 5.1, the *RMSE* of the reconstructed deformation partly reflects these short-term variations.

To clarify how different deformation patterns respond to present hydrological forcing, we analyze the periodic and time-lag relationships between detrended TS-InSAR deformation and GWL from different clusters using Short-time Fourier Transform (STFT) (Zhang et al., 2024). The detrend processing of time series allows simultaneous investigation of the aquifer response to both annual hydrological forcing and short-term extreme hydrological events.

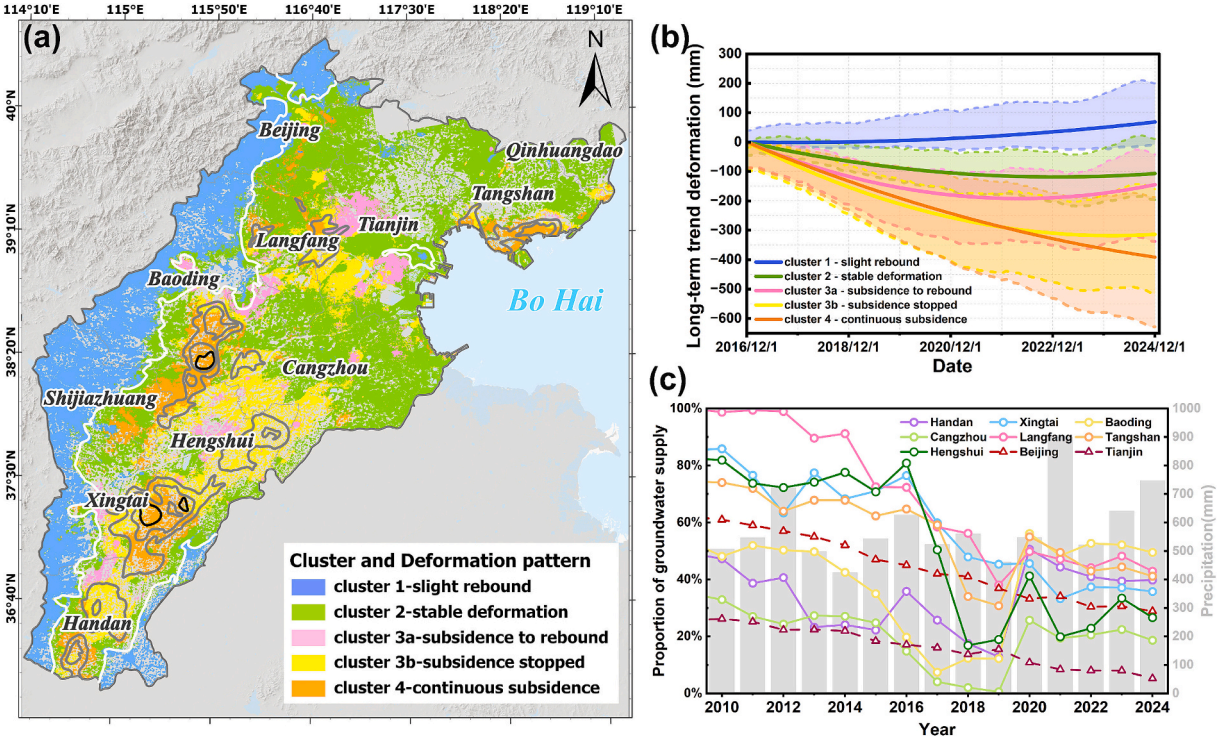

Fig. 13. Deformation pattern subdivision (cluster-3a and cluster-3b) based on $slope_{end}$ and its relationship with proportion of groundwater supply in the major cities of BTH Plain. (a) Spatial distribution. (b) Long-term deformation temporal evolution, with shaded areas indicating the variability within each cluster. (c) Temporal variations in the proportion of groundwater supply for major cities, with annual precipitation shown for comparison.

As shown in Fig. 14, GWL at all representative wells exhibit clear seasonal oscillations, with declines during summer–autumn irrigation periods and recovery during winter–spring. Correspondingly, detrended deformation time series show seasonal variability. STFT results indicate dominant resonance periods between deformation and GWL of 12–36 months, marked by red dots that exceed the 95% confidence level, suggesting that surface deformation responds to groundwater variations through a cumulative process rather than an instantaneous elastic response.

During anomalously wet years (e.g., 2021 and 2023), STFT energy intensifies markedly, and several representative wells show groundwater recovery accompanied by short-term deformation rebound. These observations suggest that the compaction-rebound behavior of the aquifer system is not only controlled by annual-scale seasonal hydrological recharge, but also by interannual precipitation variability.

Additionally, the phase difference analysis between deformation and GWL changes reveals the degree of annual lag. Wells A and B, located in cluster-1, show an annual lag < 2 months. Particularly, Well A, situated in an area dominated by elastic deformation, exhibits a very short lag of 0.35 months, suggesting that pressure diffusion and re-equilibration in the highly permeable sandy layers occur rapidly following GWL changes. In contrast, Wells E, F, G, H, and located in the significant subsidence cluster-3 and cluster-4, show substantially larger annual lags of 4–6 months. The widespread presence of low-permeability clay interbeds in these zones reduces the hydraulic conductivity of the aquifer, thereby leading to a significant lag response of deformation to annual GWL changes.

Although STFT analysis primarily captures short-term seasonal phase lags (weeks to months), such periodic responses are fundamentally distinct from the long-term subsidence delay caused by residual compaction in clay-rich aquifer systems, which is governed by slow hydraulic diffusion and delayed consolidation in aquitards, typically lasting years to decades (Hoffmann et al., 2001; Galloway and Burbey, 2011; Murray and Lohman, 2018), especially in the region of cluster-3 and cluster-4, which will be discussed in the next section.

5.4. *Control mechanism of geological structure on the long-term subsidence decay and delay process*

For cluster-3 and cluster-4 of the BTH Plain, where land subsidence has not yet ceased, we further investigate the mechanisms governing long-term delayed compaction and subsidence reduction, which represent the long-term lagged response of deformation to groundwater recovery. Long-term compaction is primarily controlled by the delayed drainage and consolidation of low-permeability clay layers within the aquifer system (Galloway et al., 1998). The aquifer system may exhibit either elastic or inelastic responses to variations in GWL, depending on whether the GWL remains above or falls below the maximum pre-consolidation head within the clay layers. Due to the prolonged history of groundwater overexploitation across the plain, the GWL has fallen far below the maximum pre-consolidation head (Yang et al., 2021). Therefore, even when the GWL begins to recover, the formed hydraulic head difference forces the adjacent clay layers to drain slowly, resulting in continued compaction (Chaussard et al., 2014).

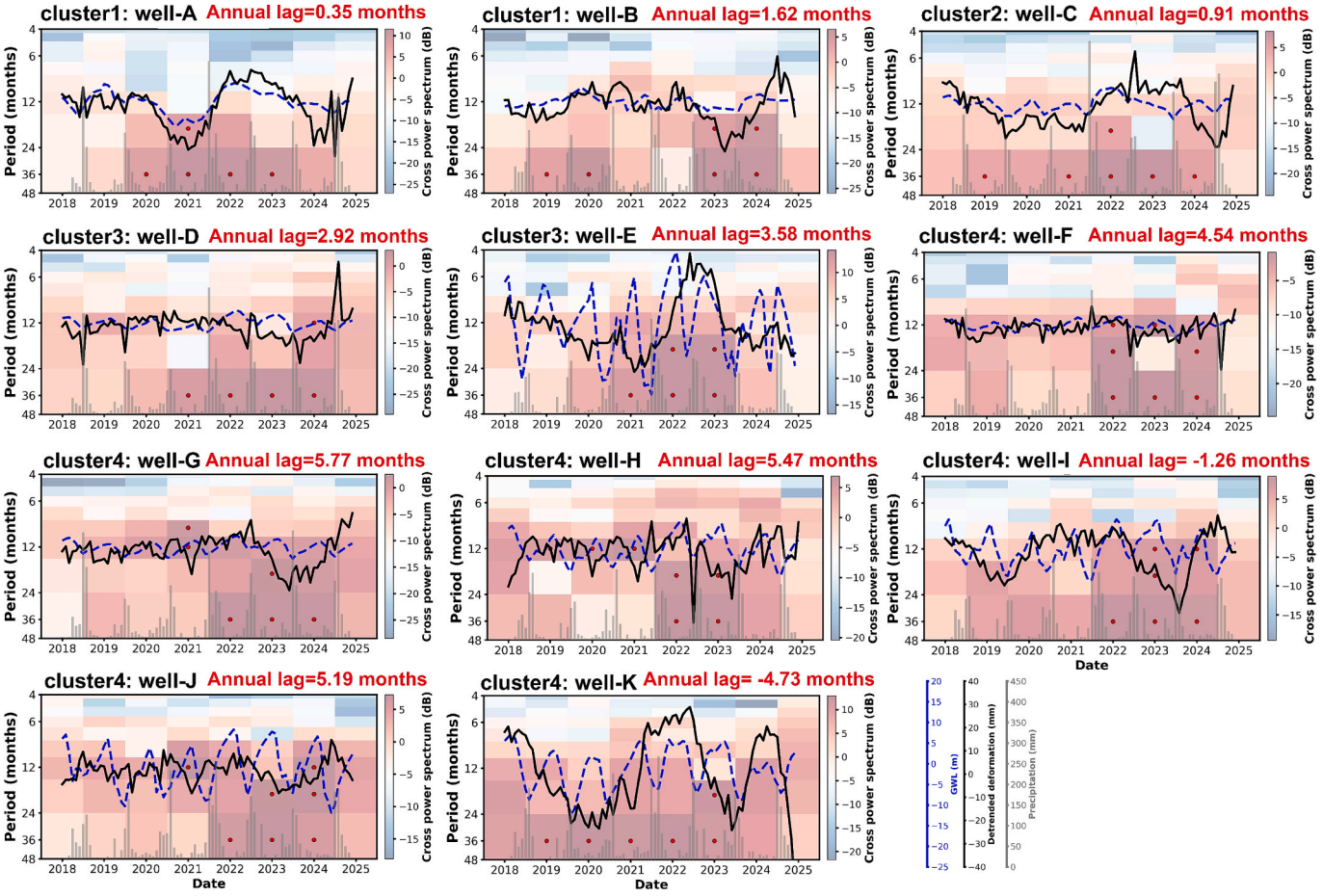

Fig. 14. Short-time cross-power spectral analysis of deformation and GWL at 11 selected wells in Fig. 9. Long-term trend was removed from the TS-InSAR time series so that only the seasonal component and the residual signal remained. The black solid line represents detrended deformation. The blue dashed line represents GWL. The red dots indicate significant responses at a 95% confidence level. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To quantify this multi-year lagged deformation response, we adopt a time-dependent exponential decay model. This is a delayed response to past mining stress. It can be represented by a time-dependent exponential decay function (Hu et al., 2018):

$$D(t) = M(1 - e^{Kt}) \tag{11}$$

where $D(t)$ is the long-term compaction component, M is the magnitude of compaction, and K is the decay coefficient. The decay coefficient K approaching 1 indicates rapid decay of subsidence, whereas K approaching 0 represents slow decay and pronounced delay. The time scale of this delayed response depends on clay thickness, vertical hydraulic conductivity, and other aquifer properties, and commonly persists for several years or even decades (Kang & Knight, 2023).

We apply this exponential model to the long-term deformation trends of cluster-3 and cluster-4 to quantify their degree of delayed compaction.

Although both cluster-3 and cluster-4 are characterized by subsidence, they exhibit distinct decay behaviors. The decay coefficients K of cluster-4 (Baoding-Xingtai) are generally lower than those of cluster-3 (Hengshui), indicating weaker attenuation of subsidence signals. Because the inverse of K can be interpreted as a characteristic timescale of subsidence decay, lower K values imply a longer persistence of delayed compaction. The observed contrast in decay behavior may be related to differences in subsurface lithological conditions. The hydrogeological profile collected in the Supplementary Materials S4 show that the Baoding–Xingtai area corresponding to cluster-4 is characterized by a thicker and more continuous clay-dominated sequence, whereas the Hengshui area (cluster-3) contains relatively thinner clay layers interbedded with silt aquifers. Thick clay layers are associated with low permeability and high compressibility, which favor slow pore-pressure dissipation and prolonged delayed consolidation. In contrast, the relatively higher K values observed in cluster-3 suggest shorter decay timescales.

Interestingly, the spatial distribution of the decay coefficient K reveals that geological structures likely influence the recovery process of the subsidence funnels. As shown in Fig. 15, low K values (<0.002) do not necessarily concentrate in the funnel centers; rather, in the Langfang, Xingtai, and coastal Tangshan funnels, low K values are concentrated near the funnel edges where faults are present. Taking the Xingtai subsidence funnel as an example, a profile along B–B' reveals that although the subsidence rate decreases from the funnel center toward the edges, the decay coefficient K simultaneously decreases, indicating that the long-term delayed compaction is actually more pronounced along the funnel edges. In contrast, in the Baoding–Shijiazhuang funnel without fault control, profile A–A' shows that low K values are concentrated in the funnel center and gradually increase toward the edges, while the subsidence rate simultaneously decreases outward. This suggests that the Baoding–Shijiazhuang funnel exhibits a more uniform radially "center to edge" recovery pattern.

In addition, from the perspective of groundwater management and land subsidence risk, the spatial variability K of attenuation coefficients is of great significance. The Low K value indicates slow subsidence attenuation and significant delay in compaction, which means that even with the continuous recovery of groundwater levels, ground subsidence may continue for many years in response to past mining stresses. These areas require long-term pumping regulation and continuous monitoring, especially in areas where geological structures exacerbate and delay drainage. In contrast, those with higher K values exhibit faster subsidence attenuation and respond promptly to groundwater management interventions. Therefore, based on this attenuation delay characteristic, it is helpful to evaluate the degree of recovery of regional ground subsidence and optimize management measures.

6. Conclusion

This study systematically reveals the spatially heterogeneous deformation evolution patterns of the Beijing-Tianjin-Hebei (BTH) Plain and their differential responses to groundwater recovery. After mosaicking the TS-InSAR deformation field, we reconstructed the seasonal and long-term deformation and further identified distinct deformation evolution patterns through developing the characteristic-constrained K-means clustering method. By integrating GWL observations, we distinguished the multi-scale response mechanisms of annual lag and long-term delay of deformation caused by GWL changes.

Our results reveal a clear spatial partitioning of deformation behaviors from 2016 to 2024, including 10.14% continuous subsidence, 18.86% reduced subsidence, 46.09% stable deformation, and 24.91% slight rebound. The clusters that have not yet stopped subsidence coincide with two major subsidence funnels, both corresponding to GWL decline funnels of different depths. These areas are still continuously affected by groundwater extraction.

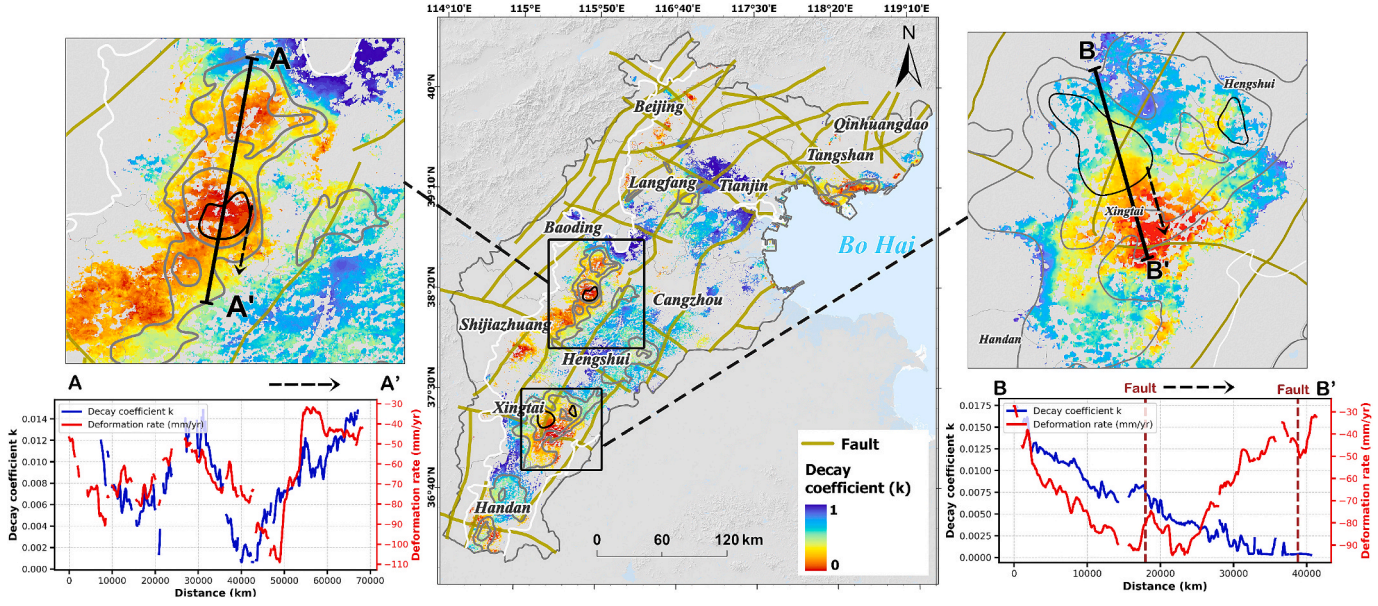

Fig. 15. The decay coefficient of the cluster-3 and cluster-4, where subsidence has not yet stopped, and its relationship with the fault distribution.

We evaluated the uncertainties in deformation reconstruction using RMSE and NRMSE. Under conditions where NRMSE remains small, RMSE indicates short-term deformation fluctuations driven by extreme hydrological events or anthropogenic disturbances. STFT analysis reveals that surface deformation exhibits varying degrees of annual-phase lag relative to GWL changes across different areas in the plain. Lag times are typically < 2 months in areas dominated by elastic deformation, but increase to 4–6 months in areas characterized by thick clay layers and significant long-term subsidence.

The exponential decay model demonstrates that the subsidence funnels recovery is controlled by geological structures. Subsidence funnels intersected by faults exhibit the slowest subsidence decay rates along their edges, indicating pronounced long-term delayed consolidation, while funnels structurally without faults show a more uniform "edge to center" radial recovery pattern.

Overall, this study establishes an integrated framework that combines TS-InSAR deformation reconstruction, deformation patterns recognition, and groundwater–deformation response analysis. The findings enhance our understanding of aquifer–land surface dynamics under emerging hydrological conditions and provide critical scientific insights for sustainable groundwater management and land subsidence mitigation in overexploited aquifer systems.

CRediT authorship contribution statement

Dexin Meng: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Beibei Chen:** Writing – review & editing, Supervision, Conceptualization. **Shubo Zhang:** Validation, Software, Investigation, Data curation. **Huili Gong:** Supervision, Conceptualization. **Chaofan Zhou:** Writing – review & editing. **Sidi Chen:** Visualization, Validation. **Kunchao Lei:** Writing – review & editing. **Haigang Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant No. 42371089, 42371081, 41930109 and 42201081) and the Beijing Outstanding Young Scientist Program (Grant No. BJJWZYJH01201910028032). The groundwater level data comes from China Institute for Geo-Environmental Monitoring (https://geocloud.cgs.gov.cn/#/).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jag.2026.105196.

Data availability

The authors do not have permission to share data.

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